EVS-Module 5 BCS508

1. E-waste refers to:

A. Organic waste
B. Waste generated from electronic and electrical devices
C. Hazardous medical waste
D. Construction waste
Answer: B

2. The E-Waste (Management) Rules, 2016, came into effect on:

A. April 1, 2016
B. October 1, 2017
C. March 15, 2015
D. August 1, 2018
Answer: A

3. Extended Producer Responsibility (EPR) under e-waste management refers to:

A. Responsibility of producers to manage waste from their products
B. Consumers segregating waste
C. Government recycling initiatives
D. Importers of electronic goods managing e-waste
Answer: A

4. The primary component of e-waste is:

A. Paper
B. Plastics and metals
C. Organic matter
D. Glass
Answer: B

5. Which of the following is a major source of e-waste?

A. Office furniture
B. Used batteries
C. Old computers and smartphones
D. Construction debris
Answer: C

6. E-waste contains hazardous substances such as:

A. Lead and mercury
B. Sodium and potassium

C. Calcium and magnesium
D. Silicon and aluminum
Answer: A

7. The concept of "urban mining" in e-waste refers to:

A. Extracting metals from landfills
B. Extracting precious materials from e-waste
C. Using e-waste for construction
D. Replacing urban landfills with recycling units
Answer: B

8. The Basel Convention relates to:

A. Regulating transboundary movements of hazardous waste
B. Guidelines for plastic waste management
C. Reduction of greenhouse gases
D. Banning open waste burning
Answer: A

9. Which country is the largest producer of e-waste globally?

A. China
B. USA
C. India
D. Japan
Answer: A

10. Which of the following is NOT a method of e-waste disposal?

A. Incineration
B. Landfilling
C. Recycling
D. Composting
Answer: D

11. Which precious metal can be recovered from e-waste?

A. Gold
B. Copper
C. Silver
D. All of the above
Answer: D

12. The main environmental risk from improper e-waste disposal is:

A. Air pollution
B. Soil and water contamination
C. Noise pollution
D. Depletion of landfills
Answer: B

13. Informal e-waste recycling is prevalent in:

A. Developed countries
B. Developing countries
C. Arctic regions
D. None of the above
Answer: B

14. The "take-back system" in e-waste management involves:

A. Consumers returning used electronics to producers
B. Recycling companies collecting e-waste
C. Governments reclaiming electronic waste
D. Municipal corporations managing e-waste
Answer: A

15. E-waste dismantling and recycling should be carried out:

A. In open spaces
B. In authorized facilities
C. By informal workers
D. At the consumer's home
Answer: B

16. Which Indian city is known for its informal e-waste recycling hubs?

A. Delhi
B. Bengaluru
C. Kolkata
D. Hyderabad
Answer: A

17. Which agency regulates e-waste management in India?

A. Ministry of Electronics and IT
B. Central Pollution Control Board (CPCB)
C. Ministry of Environment, Forest and Climate Change
D. Both B and C
Answer: D

18. What percentage of e-waste is estimated to be recycled globally?

A. 10%
B. 17%
C. 25%
D. 30%
Answer: B

19. Which element found in e-waste is highly toxic?

A. Iron
B. Mercury
C. Aluminum
D. Tin
Answer: B

20. CRT monitors contribute to e-waste through their:

A. Plastic frames
B. Leaded glass
C. Circuit boards
D. All of the above
Answer: D

21. The "Digital India" initiative in India focuses on:

A. Reducing electronic device use
B. Promoting safe e-waste recycling practices
C. Encouraging export of e-waste
D. Open dumping of e-waste
Answer: B

22. Which of the following is a challenge in e-waste management?

A. Lack of awareness among consumers
B. High costs of recycling technology
C. Informal recycling practices
D. All of the above
Answer: D

23. Lithium-ion batteries in e-waste are:

A. Harmless when disposed of in landfills
B. Flammable and hazardous
C. Composted with organic waste

D. Non-recyclable
Answer: B

24. The "Swachh Bharat Abhiyan" includes:

A. Guidelines for e-waste management
B. Ban on e-waste production
C. Restriction of e-waste import
D. All of the above
Answer: A

25. Which material is commonly recovered from printed circuit boards?

A. Wood
B. Plastic
C. Precious metals
D. Ceramics
Answer: C

26. E-waste accounts for approximately what percentage of global waste annually?

A. 1%
B. 5%
C. 10%
D. 15%
Answer: C

27. Which international organization promotes e-waste recycling?

A. World Health Organization (WHO)
B. International Telecommunication Union (ITU)
C. United Nations Environment Programme (UNEP)
D. Both B and C
Answer: D

28. Which Indian state was the first to adopt e-waste management rules?

A. Maharashtra
B. Tamil Nadu
C. Karnataka
D. Kerala
Answer: A

29. The "Right to Repair" movement is aimed at:

A. Making electronics disposable
B. Increasing the lifespan of electronic products
C. Promoting illegal recycling
D. Reducing repair options
Answer: B

30. Which of the following is a recommended practice for managing personal e-waste?

A. Open dumping
B. Donating working devices
C. Burning non-working parts
D. Mixing with general waste
Answer: B

1. E-waste is primarily composed of:

A. Organic materials
B. Plastics, metals, and glass
C. Construction debris
D. None of the above
Answer: B

2. What percentage of e-waste is made up of metals?

A. 10-15%
B. 20-25%
C. 50-60%
D. 70-80%
Answer: C

3. The largest component of e-waste by volume is:

A. Plastics
B. Circuit boards
C. Screens and monitors
D. Batteries
Answer: A

4. Which metal is commonly found in printed circuit boards of e-waste?

A. Gold
B. Zinc
C. Tin
D. Aluminum
Answer: A

5. E-waste generation is highest in which sector?

A. Industrial equipment
B. Consumer electronics
C. Medical devices
D. Agricultural tools
Answer: B

6. The fastest-growing category of e-waste is:

A. Mobile phones
B. Televisions
C. Refrigerators
D. Laptops
Answer: A

7. Which of the following devices contributes significantly to e-waste generation globally?

A. Electric vehicles
B. Smartphones and computers
C. Solar panels
D. Power transformers
Answer: B

8. What proportion of global e-waste comes from small IT and telecommunication equipment?

A. 10%
B. 20%
C. 30%
D. 50%
Answer: C

9. The main driver of e-waste generation in developed countries is:

A. Lack of awareness
B. Rapid technology obsolescence
C. High recycling costs
D. Government policies
Answer: B

10. The average lifespan of mobile phones contributing to e-waste is:

A. 2 years
B. 5 years
C. 10 years

D. 15 years
Answer: A

11. Which category of e-waste includes refrigerators and air conditioners?

A. Small household appliances
B. Large household appliances
C. IT equipment
D. None of the above
Answer: B

12. Which toxic material is often found in cathode ray tubes (CRTs)?

A. Mercury
B. Lead
C. Cadmium
D. Arsenic
Answer: B

13. How much e-waste is generated globally every year (as of recent estimates)?

A. 20 million metric tons
B. 30 million metric tons
C. 50 million metric tons
D. 70 million metric tons
Answer: C

14. What percentage of global e-waste is formally recycled?

A. 10%
B. 17%
C. 25%
D. 30%
Answer: B

15. Which of the following is a key component of e-waste from smartphones?

A. Lithium-ion batteries
B. Polyethylene plastic
C. Nickel-cadmium cells
D. None of the above
Answer: A

16. E-waste from IT equipment primarily consists of:

A. Cables
B. Circuit boards
C. Glass screens
D. All of the above
Answer: D

17. Which type of e-waste contains refrigerants harmful to the environment?

A. Mobile phones
B. Laptops
C. Refrigerators and air conditioners
D. Monitors
Answer: C

18. The highest e-waste generation per capita occurs in:

A. Asia
B. Europe
C. Africa
D. South America
Answer: B

19. The presence of rare earth metals in e-waste is significant for:

A. Electronics manufacturing
B. Fertilizer production
C. Textile industry
D. None of the above
Answer: A

20. Which country is the largest contributor to global e-waste generation?

A. India
B. USA
C. China
D. Both B and C
Answer: D

21. The e-waste generated annually in India is approximately:

A. 1.5 million metric tons
B. 2 million metric tons
C. 2.5 million metric tons
D. 3 million metric tons
Answer: B

22. Which metal is extracted from e-waste for its high conductivity?

A. Silver
B. Copper
C. Gold
D. All of the above
Answer: D

23. E-waste generation in the healthcare sector includes:

A. X-ray machines
B. MRI scanners
C. ECG monitors
D. All of the above
Answer: D

24. Obsolete televisions contribute to e-waste primarily due to:

A. Plastic frames
B. CRT glass
C. Circuit boards
D. All of the above
Answer: D

25. The percentage of plastics in e-waste is approximately:

A. 10%
B. 20%
C. 30%
D. 40%
Answer: C

26. Which category of devices contributes the least to e-waste generation?

A. Consumer electronics
B. Medical devices
C. Industrial machinery
D. None of the above
Answer: B

27. E-waste generated from solar panels is expected to:

A. Decrease due to better recycling
B. Increase due to widespread adoption
C. Remain constant

D. Have no impact on the environment
Answer: B

28. The term "planned obsolescence" in electronics refers to:

A. Designing products with a limited lifespan
B. Making durable products
C. Recycling obsolete devices
D. None of the above
Answer: A

29. Which of the following contributes most to e-waste from households?

A. Washing machines
B. Smartphones
C. Televisions
D. All of the above
Answer: D

30. The toxic gas released from burning e-waste containing plastics is:

A. Carbon dioxide
B. Hydrogen chloride
C. Dioxins and furans
D. Sulfur dioxide
Answer: C

1. The largest producer of e-waste globally is:

A. India
B. China
C. USA
D. European Union
Answer: B

2. The Basel Convention focuses on:

A. Promoting e-waste recycling
B. Regulating transboundary movement of hazardous waste
C. Reducing greenhouse gas emissions
D. Eliminating e-waste in landfills
Answer: B

3. Which region generates the highest e-waste per capita?

A. Africa
B. Asia
C. Europe
D. South America
Answer: C

4. The Global E-waste Monitor is published by:

A. UNEP
B. WHO
C. ITU
D. Both A and C
Answer: D

5. Which country has the highest formal e-waste recycling rate?

A. Germany
B. USA
C. India
D. China
Answer: A

6. Which pollutant in e-waste is highly toxic to the nervous system?

A. Lead
B. Mercury
C. Cadmium
D. Arsenic
Answer: B

7. Which of the following is considered a persistent organic pollutant in e-waste?

A. Polychlorinated biphenyls (PCBs)
B. Lead oxide
C. Aluminum oxide
D. None of the above
Answer: A

8. Informal recycling of e-waste is predominant in:

A. USA
B. Europe
C. Developing countries
D. Japan
Answer: C

9. Which of the following gases can be released during e-waste burning?

A. Carbon dioxide
B. Dioxins and furans
C. Methane
D. Hydrogen sulfide
Answer: B

10. The global annual e-waste generation is approximately:

A. 30 million metric tons
B. 50 million metric tons
C. 70 million metric tons
D. 100 million metric tons
Answer: B

11. The largest importer of illegal e-waste globally is:

A. China
B. India
C. Ghana
D. Indonesia
Answer: A

12. Polybrominated diphenyl ethers (PBDEs) in e-waste are used as:

A. Coolants
B. Flame retardants
C. Conductors
D. Insulators
Answer: B

13. The "right to repair" movement aims to:

A. Reduce e-waste generation
B. Promote consumerism
C. Ban recycling of electronics
D. Prohibit electronic repairs
Answer: A

14. Which of these is NOT a heavy metal found in e-waste?

A. Mercury
B. Lead
C. Zinc

D. Nickel
Answer: C

15. The largest contributor to e-waste pollution in landfills is:

A. Mobile phones
B. CRT monitors
C. Batteries
D. Refrigerators
Answer: B

16. What is the major environmental risk of cadmium in e-waste?

A. Groundwater contamination
B. Air pollution
C. Noise pollution
D. Deforestation
Answer: A

17. Which international framework governs the movement of hazardous e-waste?

A. Stockholm Convention
B. Kyoto Protocol
C. Basel Convention
D. Paris Agreement
Answer: C

18. Which type of e-waste is most common in developing countries?

A. Large household appliances
B. IT equipment
C. Consumer electronics
D. Small household appliances
Answer: C

19. The Stockholm Convention addresses:

A. Persistent organic pollutants in e-waste
B. Transboundary waste movement
C. E-waste legislation in developing countries
D. Recycling processes for e-waste
Answer: A

20. Which component of e-waste is linked to ozone layer depletion?

A. CRT glass
B. Plastics
C. CFCs from refrigerators
D. Lithium-ion batteries
Answer: C

21. E-waste generation in India ranks globally as:

A. First
B. Second
C. Third
D. Fifth
Answer: C

22. Which toxic element in e-waste can bioaccumulate in fish?

A. Mercury
B. Cadmium
C. Lead
D. Arsenic
Answer: A

23. One of the primary pollutants released from incinerating e-waste plastics is:

A. Sulfur dioxide
B. Dioxins
C. Nitrogen oxides
D. Ozone
Answer: B

24. Which country has implemented the "Clean India, Green Electronics" initiative?

A. China
B. India
C. Japan
D. USA
Answer: B

25. The presence of brominated flame retardants in e-waste is harmful because:

A. They release dioxins when burned
B. They are carcinogenic
C. They persist in the environment
D. All of the above
Answer: D

26. E-waste workers in informal sectors are most exposed to:

A. Lead and mercury
B. Plastics and glass
C. Noise pollution
D. Radioactive materials
Answer: A

27. The "Circular Economy" approach to e-waste focuses on:

A. Increasing landfill space
B. Promoting product lifecycle extension
C. Phasing out electronic goods
D. Restricting technology exports
Answer: B

28. E-waste recycling practices in Europe emphasize:

A. Formal recycling units
B. Extended Producer Responsibility (EPR)
C. Consumer awareness programs
D. All of the above
Answer: D

29. Which of these pollutants from e-waste is NOT a heavy metal?

A. Chromium
B. Arsenic
C. Lead
D. Hydrocarbons
Answer: D

30. Which UN Sustainable Development Goal addresses e-waste management?

A. Goal 11: Sustainable Cities and Communities
B. Goal 12: Responsible Consumption and Production
C. Goal 13: Climate Action
D. Goal 14: Life Below Water
Answer: B

1. Which of the following is a hazardous property of lead found in e-waste?

A. Causes neurological damage
B. Promotes plant growth
C. Increases water alkalinity

D. Enhances conductivity
Answer: A

2. Mercury in e-waste is hazardous because it:

A. Contaminates air
B. Bioaccumulates in the food chain
C. Causes respiratory issues
D. All of the above
Answer: D

3. Cadmium in e-waste is primarily found in:

A. Plastics
B. Batteries
C. Display screens
D. Cooling agents
Answer: B

4. Hexavalent chromium in e-waste is hazardous because:

A. It corrodes metals
B. It is a carcinogen
C. It causes ozone depletion
D. None of the above
Answer: B

5. Which of the following materials in e-waste is linked to kidney damage?

A. Mercury
B. Lead
C. Cadmium
D. Arsenic
Answer: C

6. Brominated flame retardants in e-waste are known to:

A. Persist in the environment
B. Be endocrine disruptors
C. Release toxic dioxins when burned
D. All of the above
Answer: D

7. The hazardous property of arsenic in e-waste includes:

A. Causes lung cancer
B. Bioaccumulates in aquatic systems
C. Reduces soil fertility
D. Improves metal conductivity
Answer: A

8. Which element in e-waste poses a risk of explosion when improperly handled?

A. Lithium
B. Aluminum
C. Nickel
D. Zinc
Answer: A

9. Polychlorinated biphenyls (PCBs) are hazardous because they:

A. Are non-biodegradable
B. Cause skin irritation and cancer
C. Accumulate in fat tissues
D. All of the above
Answer: D

10. Which toxic gas is released from burning e-waste containing plastics?

A. Hydrogen chloride
B. Dioxins and furans
C. Sulfur dioxide
D. Methane
Answer: B

11. E-waste hazardous properties are regulated under which international convention?

A. Basel Convention
B. Stockholm Convention
C. Rotterdam Convention
D. Kyoto Protocol
Answer: A

12. Which type of e-waste often contains hazardous polychlorinated biphenyls (PCBs)?

A. Batteries
B. Circuit boards
C. Capacitors
D. Monitors
Answer: C

13. E-waste incineration can release:

A. Lead oxides
B. Volatile organic compounds
C. Dioxins
D. All of the above
Answer: D

14. The hazardous property of lithium-ion batteries in e-waste includes:

A. Flammability
B. Corrosion of soil
C. Bioaccumulation
D. Radioactivity
Answer: A

15. Nickel in e-waste is hazardous due to its:

A. Allergic properties
B. High bioaccumulation
C. Radioactive nature
D. None of the above
Answer: A

16. Which material in e-waste can damage the central nervous system?

A. Cadmium
B. Lead
C. Mercury
D. All of the above
Answer: D

17. Hazardous property of e-waste plastics arises from:

A. Release of microplastics
B. Toxic additives like phthalates
C. Difficulty in recycling
D. Both A and B
Answer: D

18. The flammability of certain e-waste materials is due to:

A. Lithium in batteries
B. Plastics
C. Hydrocarbons in refrigerants

D. All of the above
Answer: D

19. The hazardous effects of zinc in e-waste include:

A. Groundwater contamination
B. Bioaccumulation in fish
C. Plant toxicity
D. None of the above
Answer: A

20. Which component in e-waste is linked to ozone layer depletion?

A. Chlorofluorocarbons (CFCs)
B. Lead solder
C. PCBs
D. Cadmium
Answer: A

21. Cadmium exposure from e-waste can result in:

A. Itai-itai disease
B. Bone fractures
C. Kidney failure
D. All of the above
Answer: D

22. Which of these gases can be emitted during e-waste burning?

A. Carbon monoxide
B. Hydrogen sulfide
C. Dioxins
D. Both A and C
Answer: D

23. Heavy metals in e-waste affect human health by:

A. Damaging DNA
B. Causing organ failure
C. Triggering cancer
D. All of the above
Answer: D

24. Hazardous property of cobalt in e-waste includes:

A. Radioactivity
B. Toxic to aquatic life
C. Corrosive nature
D. All of the above
Answer: B

25. Chlorinated plastics in e-waste release harmful chemicals during:

A. Recycling
B. Incineration
C. Landfilling
D. Both B and C
Answer: D

26. Hazardous property of antimony in e-waste includes:

A. Carcinogenicity
B. Flammability
C. Bioaccumulation
D. All of the above
Answer: A

27. Lead solder in e-waste is hazardous due to its:

A. Persistence in soil
B. Neurotoxicity
C. Water solubility
D. Both A and B
Answer: D

28. Arsenic in e-waste contributes to:

A. Groundwater toxicity
B. Respiratory diseases
C. Carcinogenic effects
D. All of the above
Answer: D

29. Hazardous property of PVC in e-waste includes:

A. Releases dioxins when burned
B. Biodegradable under certain conditions
C. Promotes soil fertility
D. None of the above
Answer: A

30. Hazardous exposure to beryllium in e-waste leads to:

A. Skin diseases
B. Chronic lung disease
C. Gastrointestinal issues
D. All of the above
Answer: B

1. What is e-waste?

- a) Household waste
- b) Electronic waste from old electronics
- c) Industrial waste
- d) Food waste
- **Answer: b) Electronic waste from old electronics**

2. Which of the following is NOT a common source of e-waste?

- a) Computers
- b) Mobile phones
- c) Plastic bottles
- d) Televisions
- **Answer: c) Plastic bottles**

3. Which heavy metal is commonly found in e-waste?

- a) Lead
- b) Iron
- c) Copper
- d) Aluminum
- **Answer: a) Lead**

4. How does e-waste affect human health?

- a) Causes breathing problems
- b) Leads to neurological damage
- c) Improves immune system
- d) Has no impact on health
- **Answer: b) Leads to neurological damage**

5. What is one of the major dangers of burning e-waste?

- a) Creation of greenhouse gases
- b) Release of toxic fumes like dioxins
- c) Increase in soil fertility
- d) Formation of ozone layer
- **Answer: b) Release of toxic fumes like dioxins**

6. Which of the following is a primary environmental hazard of e-waste?

- a) Water contamination
- b) Noise pollution
- c) Soil degradation
- d) Air pollution
- **Answer: a) Water contamination**

7. What does the term "e-waste recycling" refer to?

- a) Reusing old electronic products without any treatment
- b) Collecting and processing e-waste to recover valuable materials
- c) Disposing of e-waste in landfills
- d) Burning electronic waste to reduce its volume
- **Answer: b) Collecting and processing e-waste to recover valuable materials**

8. Which of the following is a toxic substance found in e-waste?

- a) Cadmium
- b) Sodium
- c) Potassium
- d) Magnesium
- **Answer: a) Cadmium**

9. Which of the following health issues is linked to prolonged exposure to e-waste pollutants?

- a) Skin infections
- b) Cancer
- c) Eye strain
- d) Asthma
- **Answer: b) Cancer**

10. What is the effect of lead exposure from e-waste on humans?

- a) Improves cognitive function
- b) Causes kidney damage
- c) Prevents muscle weakness
- d) Enhances vision
- **Answer: b) Causes kidney damage**

11. How does e-waste affect aquatic life?

- a) Increases oxygen levels
- b) Causes the death of marine species
- c) Reduces the temperature of water bodies
- d) Enhances aquatic plant growth

- **Answer: b) Causes the death of marine species**

12. What is a major cause of air pollution from e-waste?

- a) Burning of plastic components
- b) Release of oxygen gas
- c) Emission of water vapor
- d) Release of CO2 from batteries
- **Answer: a) Burning of plastic components**

13. Which country is known for being the largest producer of e-waste?

- a) India
- b) United States
- c) China
- d) Brazil
- **Answer: b) United States**

14. Which of the following is NOT a recommended method of e-waste disposal?

- a) Recycling
- b) Landfilling
- c) Incineration
- d) Reuse
- **Answer: b) Landfilling**

15. Which part of e-waste can be recycled for reuse in new products?

- a) Printed Circuit Boards (PCBs)
- b) Plastic casings only
- c) Batteries only
- d) Liquid components
- **Answer: a) Printed Circuit Boards (PCBs)**

16. Which electronic component contains mercury that can harm the environment?

- a) Computer monitors
- b) Lead-acid batteries
- c) Refrigerator compressors
- d) Plastic casings
- **Answer: a) Computer monitors**

17. Which of the following is an example of a material extracted from e-waste during recycling?

- a) Gold
- b) Sodium chloride

- c) Silicon dioxide
- d) Hydrogen gas
- **Answer: a) Gold**

18. Which of the following is a risk of improper disposal of e-waste?

- a) Creation of valuable resources
- b) Soil and water contamination
- c) Increased employment opportunities
- d) Decrease in carbon emissions
- **Answer: b) Soil and water contamination**

19. What are the primary impacts of e-waste on children's health?

- a) Lead poisoning and developmental delays
- b) Improved immune system
- c) Increased height and weight
- d) Enhanced brain development
- **Answer: a) Lead poisoning and developmental delays**

20. Which material in e-waste is commonly recycled to recover valuable metals?

- a) Plastic
- b) Glass
- c) Copper
- d) Silicon
- **Answer: c) Copper**

21. What is the role of the Basel Convention regarding e-waste?

- a) Promoting e-waste collection
- b) Banning international trade of hazardous e-waste
- c) Promoting e-waste recycling in industrial countries
- d) Increasing the production of e-waste
- **Answer: b) Banning international trade of hazardous e-waste**

22. Which of the following is a major environmental impact of e-waste dumping in landfills?

- a) Reduction in soil fertility
- b) Release of toxic chemicals into the soil
- c) Increase in biodiversity
- d) Decrease in water contamination
- **Answer: b) Release of toxic chemicals into the soil**

23. Which of the following materials is commonly found in batteries that are part of e-waste?

- a) Lithium

- b) Lead
- c) Zinc
- d) All of the above
- **Answer: d) All of the above**

24. What is one way to reduce e-waste?

- a) Extending the life cycle of electronics through repair and reuse
- b) Burning more e-waste
- c) Using more single-use electronics
- d) Disposing e-waste in landfills
- **Answer: a) Extending the life cycle of electronics through repair and reuse**

25. What happens when e-waste is improperly incinerated?

- a) It produces harmless gas
- b) It creates toxic chemicals like dioxins and furans
- c) It increases the recyclability of materials
- d) It reduces global warming
- **Answer: b) It creates toxic chemicals like dioxins and furans**

26. Which chemical found in e-waste can cause lung damage when inhaled?

- a) Arsenic
- b) Lead
- c) Cadmium
- d) Mercury
- **Answer: c) Cadmium**

27. Which of the following is a safe way to manage e-waste?

- a) Collecting and recycling it through certified e-waste recyclers
- b) Burning it in open areas
- c) Storing it in landfills
- d) Dumping it in water bodies
- **Answer: a) Collecting and recycling it through certified e-waste recyclers**

28. What is the impact of e-waste on the ozone layer?

- a) It does not affect the ozone layer
- b) It depletes the ozone layer
- c) It helps to restore the ozone layer
- d) It thickens the ozone layer
- **Answer: b) It depletes the ozone layer**

29. What is the primary function of the informal e-waste sector in many developing countries?

- a) Safe disposal of e-waste
- b) High-efficiency recycling of e-waste
- c) Informal recycling, often without proper safety measures
- d) Exportation of e-waste to developed countries
- **Answer: c) Informal recycling, often without proper safety measures**

30. Which strategy is most effective in reducing the impact of e-waste on health and the environment?

- a) Improper disposal in landfills
- b) Recycling through formal, regulated channels
- c) Burning e-waste to reduce volume
- d) Reusing old electronics without recycling

Answer: b) Recycling through formal, regulated channels

1. What does e-waste refer to?

- a) Only old computers
- b) Any waste from used or broken electronics
- c) Food waste from electronics factories
- d) Water waste from factories
- **Answer: b) Any waste from used or broken electronics**

2. Which of the following is NOT considered e-waste?

- a) Old televisions
- b) Broken mobile phones
- c) Empty plastic bottles
- d) Outdated computers
- **Answer: c) Empty plastic bottles**

3. What is the first step in managing domestic e-waste?

- a) Incinerating e-waste
- b) Sorting and separating e-waste
- c) Dumping e-waste in landfills
- d) Burning e-waste to reduce volume
- **Answer: b) Sorting and separating e-waste**

4. Which of the following is a key principle of e-waste management?

- a) Maximize waste disposal

- b) Reduce, reuse, and recycle
- c) Burn e-waste to reduce its volume
- d) Export all e-waste to other countries
- **Answer: b) Reduce, reuse, and recycle**

5. Which of these is an effective method for domestic e-waste disposal?

- a) Throwing electronics in the trash
- b) Selling electronics without checking their condition
- c) Recycling e-waste at certified collection points
- d) Burning electronics in an open space
- **Answer: c) Recycling e-waste at certified collection points**

6. Which type of waste can be hazardous in e-waste?

- a) Plastic
- b) Metals like lead, mercury, and cadmium
- c) Paper
- d) Cardboard
- **Answer: b) Metals like lead, mercury, and cadmium**

7. What should NOT be done with domestic e-waste?

- a) Recycle at certified facilities
- b) Store in your home for future use
- c) Donate usable items to others
- d) Dispose of it in an environmentally responsible manner
- **Answer: b) Store in your home for future use**

8. Which of the following is a risk associated with improper e-waste disposal?

- a) Creation of valuable resources
- b) Soil and water contamination
- c) Decrease in carbon emissions
- d) Improvement in soil quality
- **Answer: b) Soil and water contamination**

9. Which material commonly found in e-waste can cause serious harm if not disposed of properly?

- a) Copper
- b) Glass
- c) Lead
- d) Wood
- **Answer: c) Lead**

10. Which principle is part of the 'reduce' strategy in e-waste management?

- a) Increasing the use of single-use electronics
- b) Minimizing the production of e-waste through conscious consumption
- c) Exporting e-waste to other countries
- d) Burning e-waste to decrease its volume
- **Answer: b) Minimizing the production of e-waste through conscious consumption**

11. What is the impact of improperly recycling e-waste at informal facilities?

- a) Protection of the environment
- b) Release of toxic substances into the environment
- c) Reduction of waste materials
- d) Better recovery of valuable metals
- **Answer: b) Release of toxic substances into the environment**

12. Which organization regulates the international movement of hazardous e-waste?

- a) WHO (World Health Organization)
- b) Basel Convention
- c) WTO (World Trade Organization)
- d) UNEP (United Nations Environment Programme)
- **Answer: b) Basel Convention**

13. Which of these should be done when disposing of domestic e-waste?

- a) Contact local e-waste recycling centers
- b) Throw it in the regular trash
- c) Burn it to reduce its volume
- d) Dismantle and reuse parts without any safety measures
- **Answer: a) Contact local e-waste recycling centers**

14. How can individuals minimize e-waste in their homes?

- a) By buying new electronics frequently
- b) By repairing and reusing older electronics
- c) By throwing away broken electronics immediately
- d) By exporting used electronics to other countries
- **Answer: b) By repairing and reusing older electronics**

15. Which of these is a benefit of e-waste recycling?

- a) Reduced resource recovery
- b) Increased environmental pollution
- c) Recovery of valuable materials like gold and copper
- d) Increased carbon emissions
- **Answer: c) Recovery of valuable materials like gold and copper**

16. What type of e-waste can often be reused instead of disposed of?

- a) Broken television screens
- b) Old batteries
- c) Working computers and mobile phones
- d) Non-functional parts
- **Answer: c) Working computers and mobile phones**

17. What is the principle of 'reuse' in e-waste management?

- a) Recycle materials only once
- b) Repair and recondition old electronics for further use
- c) Dispose of all electronics immediately
- d) Incinerate electronics to eliminate waste
- **Answer: b) Repair and recondition old electronics for further use**

18. What is the most common way to manage e-waste in households?

- a) Throw away old devices in the trash
- b) Hand old devices to electronic repair shops
- c) Collect and store all old electronics
- d) Sell or donate used but functional electronics
- **Answer: d) Sell or donate used but functional electronics**

19. Which of the following is the correct approach for disposing of e-waste containing toxic substances?

- a) Dump in a landfill
- b) Send to a certified e-waste recycler
- c) Leave it in your home
- d) Incinerate it in an open field
- **Answer: b) Send to a certified e-waste recycler**

20. Which of these metals is commonly found in e-waste and poses a risk to human health?

- a) Magnesium
- b) Iron
- c) Mercury
- d) Sodium
- **Answer: c) Mercury**

21. What does the "reduce" strategy in e-waste management emphasize?

- a) Reducing the consumption of electronics
- b) Increasing the production of electronics
- c) Burning electronics to reduce waste
- d) Dumping electronics in landfills

- **Answer: a) Reducing the consumption of electronics**

22. How are household batteries typically disposed of in e-waste management?

- a) In the regular trash
- b) By burying them underground
- c) At specialized collection points for recycling
- d) By burning them
- **Answer: c) At specialized collection points for recycling**

23. Which is a key benefit of e-waste recycling?

- a) It pollutes the environment
- b) It recovers valuable metals like gold and silver
- c) It increases energy consumption
- d) It produces harmful gases
- **Answer: b) It recovers valuable metals like gold and silver**

24. What is one of the dangers of open burning of e-waste?

- a) It produces clean energy
- b) It releases harmful chemicals like dioxins and furans
- c) It decreases air pollution
- d) It improves the quality of soil
- **Answer: b) It releases harmful chemicals like dioxins and furans**

25. What should consumers do with old, non-functional electronics?

- a) Keep them stored indefinitely
- b) Recycle them at an e-waste recycling center
- c) Burn them for disposal
- d) Throw them in the regular trash
- **Answer: b) Recycle them at an e-waste recycling center**

26. What does 'design for environment' mean in the context of e-waste management?

- a) Designing electronics that are easier to repair and recycle
- b) Designing electronics to be disposable
- c) Designing electronics with more harmful chemicals
- d) Designing electronics with shorter life cycles
- **Answer: a) Designing electronics that are easier to repair and recycle**

27. What is one of the environmental impacts of e-waste in landfills?

- a) It promotes biodiversity
- b) It causes soil and water contamination with heavy metals
- c) It improves the quality of air

- d) It increases agricultural productivity
- **Answer: b) It causes soil and water contamination with heavy metals**

28. What is a sustainable practice for e-waste management at the household level?

- a) Replace old electronics with new ones frequently
- b) Use electronics until they no longer function and then recycle them
- c) Discard electronics in regular trash bins
- d) Throw electronics away in water bodies
- **Answer: b) Use electronics until they no longer function and then recycle them**

29. What role do consumers play in e-waste management?

- a) Encouraging others to throw away electronics
- b) Recycling and properly disposing of old electronics
- c) Ignoring the problem and letting it grow
- d) Burning old electronics to save space
- **Answer: b) Recycling and properly disposing of old electronics**

30. Which of the following is a major environmental concern of e-waste disposal in developing countries?

- a) The cost of recycling
- b) The large number of valuable materials recovered
- c) The informal handling and unsafe disposal methods
- d) The production of new electronics
- **Answer: c) The informal handling and unsafe disposal methods**

1. What is the main objective of E-waste management?

- a) Increase e-waste generation
- b) Reduce environmental pollution
- c) Discard all electronic waste in landfills
- d) Store e-waste in landfills
- **Answer: b) Reduce environmental pollution**

2. Which of the following is a component of e-waste management?

- a) Disposal in landfills
- b) Collection and recycling of e-waste
- c) Burning e-waste to reduce volume
- d) Ignoring the hazardous impact of e-waste
- **Answer: b) Collection and recycling of e-waste**

3. Which of these is NOT included as part of the e-waste management system?

- a) Collection
- b) Transportation
- c) Disposal in open fields
- d) Recycling
- **Answer: c) Disposal in open fields**

4. What does the "E-waste (Management and Handling) Rules, 2011" primarily regulate?

- a) Recycling of plastic waste
- b) Handling and recycling of electronic waste
- c) Domestic waste disposal
- d) Carbon emissions from industries
- **Answer: b) Handling and recycling of electronic waste**

5. Under the E-waste Rules, which entity is responsible for the collection of e-waste?

- a) Households
- b) Local authorities
- c) Producers/Manufacturers
- d) Environmental NGOs
- **Answer: c) Producers/Manufacturers**

6. Which material in e-waste is harmful to human health and requires special disposal?

- a) Copper
- b) Glass
- c) Mercury
- d) Plastic
- **Answer: c) Mercury**

7. Who is responsible for ensuring proper disposal of e-waste under the E-waste Rules?

- a) Consumers only
- b) Government agencies only
- c) Producers, consumers, and recyclers
- d) Only the local municipal authorities
- **Answer: c) Producers, consumers, and recyclers**

8. Which of the following is part of the extended producer responsibility (EPR) under the E-waste Rules?

- a) Producers are responsible for the collection and recycling of their products
- b) Consumers must burn their e-waste

- c) Local authorities must store e-waste
- d) Importers of electronic products manage e-waste in landfills
- **Answer: a) Producers are responsible for the collection and recycling of their products**

9. The E-waste (Management and Handling) Rules came into force in which year?

- a) 2007
- b) 2011
- c) 2015
- d) 2020
- **Answer: b) 2011**

10. What is the role of authorized recyclers under the E-waste Rules?

- a) They dispose of e-waste in landfills
- b) They collect and recycle e-waste responsibly
- c) They manufacture new electronics from e-waste
- d) They burn e-waste to reduce volume
- **Answer: b) They collect and recycle e-waste responsibly**

11. Which of these components are critical in e-waste management?

- a) Collection, transportation, and recycling
- b) Ignoring environmental guidelines
- c) Exporting e-waste to other countries
- d) Storing e-waste in open areas
- **Answer: a) Collection, transportation, and recycling**

12. Which of these is a potential environmental hazard of e-waste?

- a) Soil and water contamination
- b) Increase in plant growth
- c) Reduction in greenhouse gases
- d) Promotion of biodiversity
- **Answer: a) Soil and water contamination**

13. Who is responsible for the safe disposal of e-waste at the consumer level?

- a) Only government authorities
- b) Only the e-waste manufacturers
- c) Consumers are responsible for the safe disposal
- d) Environmental agencies
- **Answer: c) Consumers are responsible for the safe disposal**

14. Which of the following is a prohibited method of e-waste disposal under the E-waste Rules?

- a) Sending e-waste to certified recycling centers
- b) Burning e-waste in open spaces
- c) Reusing components in new electronics
- d) Safe collection by producers
- **Answer: b) Burning e-waste in open spaces**

15. Which of these is included in the implementation of the E-waste Rules?

- a) A ban on the import of e-waste
- b) A ban on the sale of electronics
- c) Establishment of collection centers for e-waste
- d) Elimination of all recycling practices
- **Answer: c) Establishment of collection centers for e-waste**

16. Which of the following does not qualify as an e-waste management activity?

- a) Reprocessing and recovering materials from e-waste
- b) Creating awareness about proper disposal
- c) Storing e-waste in an open field
- d) Transporting e-waste to authorized recyclers
- **Answer: c) Storing e-waste in an open field**

17. Under the E-waste Rules, who is responsible for creating awareness about e-waste disposal?

- a) Only government agencies
- b) Manufacturers and producers
- c) Only consumers
- d) International organizations
- **Answer: b) Manufacturers and producers**

18. Which of the following must e-waste management facilities be authorized to do under the E-waste Rules?

- a) Incinerate e-waste
- b) Recycle and dispose of e-waste safely
- c) Export e-waste to foreign countries
- d) Store e-waste indefinitely
- **Answer: b) Recycle and dispose of e-waste safely**

19. Which of the following is an important principle of E-waste management?

- a) E-waste should be burned to reduce its volume

- b) E-waste should be stored in landfills without treatment
- c) E-waste should be recycled in an environmentally friendly manner
- d) E-waste should be exported without regulation
- **Answer: c) E-waste should be recycled in an environmentally friendly manner**

20. Which hazardous material found in e-waste can lead to kidney damage if improperly handled?

- a) Cadmium
- b) Copper
- c) Gold
- d) Iron
- **Answer: a) Cadmium**

21. What is the main aim of Extended Producer Responsibility (EPR) under the E-waste Rules?

- a) To reduce the cost of e-waste recycling
- b) To increase the collection and recycling of e-waste by producers
- c) To encourage consumers to keep using old electronics
- d) To reduce the import of electronics
- **Answer: b) To increase the collection and recycling of e-waste by producers**

22. Which organization is responsible for monitoring the implementation of the E-waste Rules in India?

- a) Ministry of Environment, Forest and Climate Change (MoEFCC)
- b) World Health Organization (WHO)
- c) Central Pollution Control Board (CPCB)
- d) Greenpeace
- **Answer: c) Central Pollution Control Board (CPCB)**

23. What is the main focus of the E-waste (Management and Handling) Rules, 2011?

- a) Regulating e-waste disposal in international markets
- b) Preventing the generation of e-waste
- c) Promoting responsible handling, collection, and recycling of e-waste
- d) Encouraging the use of more electronics
- **Answer: c) Promoting responsible handling, collection, and recycling of e-waste**

24. What is the requirement for producers under the E-waste Rules?

- a) Provide a waste collection scheme for their products
- b) Incinerate their products
- c) Stop producing electronics
- d) Sell electronics to only one retailer

- **Answer: a) Provide a waste collection scheme for their products**

25. What does the 'Handling' in E-waste (Management and Handling) Rules primarily refer to?

- a) The disposal of e-waste in incinerators
- b) The recovery of metals like gold from e-waste
- c) The collection, storage, transportation, and recycling of e-waste
- d) The export of e-waste to other countries

Answer: c) The collection, storage, transportation, and recycling of e-waste

1. What is the primary aim of the E-waste (Management) Rules, 2022?

- a) To promote e-waste exports
- b) To regulate the import of e-waste
- c) To ensure safe disposal and recycling of e-waste
- d) To ban the use of electronics
- **Answer: c) To ensure safe disposal and recycling of e-waste**

2. Under the E-waste (Management) Rules, 2022, who is responsible for the collection and recycling of e-waste?

- a) Only the consumers
- b) Only the government
- c) Producers, consumers, and recyclers
- d) Only local municipal authorities
- **Answer: c) Producers, consumers, and recyclers**

3. What does Extended Producer Responsibility (EPR) mean in the E-waste (Management) Rules, 2022?

- a) Producers must pay for the disposal of e-waste
- b) Producers must collect and recycle e-waste generated from their products
- c) Consumers are responsible for collecting e-waste
- d) Producers are not involved in e-waste management
- **Answer: b) Producers must collect and recycle e-waste generated from their products**

4. Which of the following is a significant feature of the E-waste (Management) Rules, 2022?

- a) Prohibition on the recycling of e-waste
- b) Inclusion of the concept of "EPR Authorization"
- c) Ban on the use of electronics
- d) Allowing uncontrolled disposal of e-waste
- **Answer: b) Inclusion of the concept of "EPR Authorization"**

5. Under the 2022 E-waste Rules, which new category has been introduced for producers to manage?

- a) Large-scale producers only
- b) Producers of products that generate hazardous e-waste
- c) Producers of electrical and electronic equipment
- d) Recyclers only
- **Answer: c) Producers of electrical and electronic equipment**

6. Which of the following is an implication of the E-waste (Management) Rules, 2022?

- a) Reduction in the responsibility of manufacturers
- b) Empowerment of producers for effective waste management
- c) Unlimited imports of electronic waste
- d) No involvement of the consumer in e-waste management
- **Answer: b) Empowerment of producers for effective waste management**

7. What is the main objective of introducing "EPR authorization" in the 2022 rules?

- a) To allow more informal e-waste recycling
- b) To regulate and monitor the collection and recycling of e-waste by producers
- c) To reduce the production of e-waste
- d) To encourage disposal of e-waste in landfills
- **Answer: b) To regulate and monitor the collection and recycling of e-waste by producers**

8. Which of the following is included in the E-waste (Management) Rules, 2022 as a new provision for manufacturers?

- a) Manufacturers are required to implement awareness programs for consumers
- b) Producers can import e-waste without restrictions
- c) E-waste management is solely the responsibility of consumers
- d) Producers are allowed to discard e-waste in landfills
- **Answer: a) Manufacturers are required to implement awareness programs for consumers**

9. What does the "Material Recovery Rate" refer to in the context of the E-waste Rules, 2022?

- a) The total quantity of e-waste produced in a year
- b) The percentage of material recovered from e-waste for reuse
- c) The energy consumed in e-waste recycling
- d) The amount of electronic devices sold each year
- **Answer: b) The percentage of material recovered from e-waste for reuse**

10. Which authority is responsible for enforcing the E-waste (Management) Rules, 2022 in India?

- a) Ministry of Environment, Forest, and Climate Change (MoEFCC)
- b) Central Pollution Control Board (CPCB)
- c) Department of Electronics and Information Technology (DeitY)
- d) Ministry of Finance
- **Answer: b) Central Pollution Control Board (CPCB)**

11. What is the key focus of the E-waste (Management) Rules, 2022 regarding the handling of hazardous components in e-waste?

- a) Allowing uncontrolled disposal of hazardous components
- b) Strict regulations for handling and disposal of hazardous components
- c) Encouraging the burning of hazardous components
- d) Removing the responsibility of producers for hazardous components
- **Answer: b) Strict regulations for handling and disposal of hazardous components**

12. What does the E-waste (Management) Rules, 2022 require producers to do in terms of collection targets?

- a) Producers must meet specific collection and recycling targets set by the government
- b) Producers are not required to meet any collection targets
- c) Producers must only collect e-waste from rural areas
- d) Producers can ignore collection targets if they donate electronics
- **Answer: a) Producers must meet specific collection and recycling targets set by the government**

13. Under the E-waste (Management) Rules, 2022, who is responsible for the safe disposal of hazardous e-waste materials?

- a) Only local authorities
- b) Producers and authorized recyclers
- c) Consumers only
- d) Government only
- **Answer: b) Producers and authorized recyclers**

14. What is the role of registered recyclers in the E-waste (Management) Rules, 2022?

- a) To sell e-waste to other countries
- b) To store e-waste indefinitely
- c) To recycle e-waste and recover valuable materials
- d) To burn e-waste to reduce volume
- **Answer: c) To recycle e-waste and recover valuable materials**

15. What are the producers required to do under the "E-waste Management and Handling" framework in the 2022 rules?

- a) Only focus on product sales
- b) Create take-back systems and ensure the safe recycling of e-waste
- c) Only focus on consumer awareness
- d) Ignore the environmental impact of their products
- **Answer: b) Create take-back systems and ensure the safe recycling of e-waste**

16. What new feature is introduced in the E-waste (Management) Rules, 2022 regarding the tracking of e-waste?

- a) No tracking is required
- b) Introduction of an online system for tracking e-waste from collection to recycling
- c) E-waste tracking is done manually at the local level
- d) E-waste tracking is only required for international shipments
- **Answer: b) Introduction of an online system for tracking e-waste from collection to recycling**

17. Under the E-waste (Management) Rules, 2022, which of the following is an obligation of producers?

- a) Recycling e-waste free of charge
- b) Providing incentives to consumers for proper disposal
- c) Paying fines for non-compliance
- d) Developing and implementing an EPR plan for their products
- **Answer: d) Developing and implementing an EPR plan for their products**

18. Which e-waste-related aspect does the 2022 Rules emphasize for better environmental protection?

- a) Increasing the use of non-recyclable materials in electronics
- b) Reducing the overall generation of e-waste through design changes
- c) Encouraging burning of e-waste
- d) Ignoring the environmental effects of e-waste disposal
- **Answer: b) Reducing the overall generation of e-waste through design changes**

19. Which of the following groups is NOT directly involved in e-waste management under the 2022 rules?

- a) Manufacturers and producers
- b) Consumers
- c) Unauthorized recycling centers
- d) Authorized recyclers
- **Answer: c) Unauthorized recycling centers**

20. The E-waste (Management) Rules, 2022 aim to minimize which of the following environmental hazards?

- a) Air pollution from burning e-waste
- b) Water contamination due to improper disposal
- c) Soil degradation from hazardous chemicals in e-waste
- d) All of the above
- **Answer: d) All of the above**

21. Which of these is a responsibility of consumers under the E-waste (Management) Rules, 2022?

- a) Collect and recycle e-waste at their own expense
- b) Dispose of e-waste in landfills
- c) Hand over e-waste to authorized collection centers or producers
- d) Sell e-waste to unauthorized recyclers
- **Answer: c) Hand over e-waste to authorized collection centers or producers**

22. How does the E-waste (Management) Rules, 2022 help in reducing the informal recycling of e-waste?

- a) By regulating and promoting authorized recycling processes
- b) By allowing unregulated collection and disposal of e-waste
- c) By encouraging the burning of e-waste
- d) By eliminating the need for recycling altogether
- **Answer: a) By regulating and promoting authorized recycling processes**

23. Which of the following is an example of a product category covered by the E-waste (Management) Rules, 2022?

- a) Kitchen appliances
- b) Refrigerators
- c) Only mobile phones
- d) All electrical and electronic equipment
- **Answer: d) All electrical and electronic equipment**

24. What is the expected result of proper implementation of the E-waste (Management) Rules, 2022?

- a) Increased generation of e-waste
- b) Reduction in the environmental and health impacts of e-waste
- c) Decrease in the cost of electronic products
- d) Reduction in the lifespan of electronic products
- **Answer: b) Reduction in the environmental and health impacts of e-waste**

25. What is the role of the Central Pollution Control Board (CPCB) under the E-waste (Management) Rules, 2022?

- a) To promote the sale of electronic products
- b) To provide EPR authorization and monitor compliance
- c) To allow unregulated disposal of e-waste
- d) To reduce electronic production
- **Answer: b) To provide EPR authorization and monitor compliance**

26. What does the term "EPR plan" refer to under the E-waste (Management) Rules, 2022?

- a) A consumer education plan
- b) A plan for the safe disposal of electronics
- c) A strategy by producers to collect, recycle, and manage e-waste
- d) A financial plan for selling electronics
- **Answer: c) A strategy by producers to collect, recycle, and manage e-waste**

27. Which of the following is required from producers in terms of the take-back system under the E-waste (Management) Rules, 2022?

- a) Producers must collect and recycle e-waste from consumers
- b) Producers can ignore take-back responsibilities
- c) Consumers must store e-waste indefinitely
- d) Producers must burn e-waste to reduce volume
- **Answer: a) Producers must collect and recycle e-waste from consumers**

28. How does the E-waste (Management) Rules, 2022 address international trade in e-waste?

- a) By banning the import and export of e-waste
- b) By allowing the free movement of e-waste across borders
- c) By regulating the import and export of e-waste to ensure safe disposal
- d) By encouraging the export of e-waste to developing countries
- **Answer: c) By regulating the import and export of e-waste to ensure safe disposal**

29. Which of the following is NOT a target under the E-waste (Management) Rules, 2022 for producers?

- a) Collection of e-waste from consumers
- b) Setting up recycling plants
- c) Ignoring the environmental impact of e-waste
- d) Achieving a certain recovery rate of materials from e-waste
- **Answer: c) Ignoring the environmental impact of e-waste**

30. What does the introduction of "E-waste mapping" aim to achieve in the E-waste (Management) Rules, 2022?

- a) Mapping e-waste globally
- b) Identifying and tracking the sources and flows of e-waste for better management
- c) Reducing the volume of e-waste generated
- d) Increasing informal recycling activities
- **Answer: b) Identifying and tracking the sources and flows of e-waste for better management**